

Assessment: UBA Kenya Retail Credit Early-Warning Model

Business context

UBA Kenya wants an early warning model that flags customers likely to default in the next 90 days so the bank can:

- prioritize collections / relationship manager outreach,
- adjust limits for revolving products,
- offer restructuring early (where appropriate).

Decision constraint: predictions must be based on data known **as of snapshot_date**.

Objective: maximize risk capture at limited capacity.

- The collections team can contact only the **top 5% highest-risk** snapshots each month.

What you have in the zip file

- uba_kenya_candidate_train.csv (labeled)
- uba_kenya_candidate_scoring.csv (unlabeled)
- DATA_DICTIONARY.md
- Instructions below

Tasks

1. **Build a baseline + one improved model**
 - Predict default_next_90d using only valid “as-of snapshot” information.
 - Provide a simple, reproducible pipeline (notebook or scripts).
2. **Validation design (must be time-aware)**
 - Use a **time-based split** (example rule you can specify): Train on snapshots before 2025-11-01, validate on snapshots on/after 2025-11-01.
 - Also ensure you are not accidentally leaking by customer duplication (customers appear multiple times).
3. **Leakage & data issues report**
 - Identify **at least 3** data issues and show evidence.
 - Specifically: call out any columns you believe are leakage and why.
4. **Operational recommendation**
 - For the validation period:

- compute **AUC** (or PR-AUC), and **Recall@Top5%** (how many defaulters captured in top 5% risk).
 - choose a threshold for “top 5%” and explain the tradeoff.
 - Provide a short strategy note:
 - “If we contact the top 5%, here's what we expect to catch.”
 - Mention at least 2 failure modes and how you'd monitor drift.
5. **Generate predictions**
- Output predictions.csv with columns:
 - customer_id, snapshot_date, pred_default_next_90d (probability)

Deliverables

- Code (notebook or python scripts)
- README.md (1–2 pages max)
- predictions.csv